

CTNT 2026 - Elliptic Curve Cryptography
Suggested Exercises

1. Find the least non-negative residue class of $6^{182} \bmod 13$.
2. Find the least non-negative residue class of $6^{-58} \bmod 13$.
3. Show that 6 is a primitive root modulo 13.
4. Find x such that $6^x \equiv 5 \bmod 13$.
5. Find the multiplicative order of 2 modulo 4003.
6. Make a table of primes (up to a bound) for which 2 is a primitive root. What kind of statistics do you notice?
7. Find a natural number $b \geq 1$ such that $65^b \equiv 2 \bmod 4003$.
8. Show that $y = x^3$ is not an elliptic curve.
9. Let $E/\mathbb{Q}$ be the elliptic curve $y^2 = x^3 - 2$ and let $P = (3, 5)$. Use the geometric definition of the addition on E to find the coordinates of $2P$ and $3P$.
10. Let E be the elliptic curve given by $y^2 = x^3 + 1$ over $\mathbb{F}_7$. Find all the points on E over $\mathbb{F}_7$ and describe the group structure of $E(\mathbb{F}_7)$.
11. Let p be a prime and let $E : y^2 = x^3 + 4x + 6$.
 - (a) For what primes p is the curve E an elliptic curve?
 - (b) For $p = 3, 5, 7, 11, 13, 17, 19, 4001, 4003$, find the number of points on $E(\mathbb{F}_p)$, the difference $(p + 1) - \#E(\mathbb{F}_p)$, and $2\sqrt{p}$. Verify that the Hasse bound holds in each case.
 - (c) Show that $E(\mathbb{F}_{4001})$ is a cyclic group.
12. Use Shank's baby-step giant-step algorithm to find x such that

$$5^x \equiv 3 \bmod 23.$$
 - (a) The order of 5 mod 23 is 22, so compute $n = 1 + \lfloor \sqrt{22} \rfloor$.
 - (b) Let $g = 5$. Compute List 1: $1, g, g^2, \dots, g^n \bmod 23$.
 - (c) Compute $u \equiv g^{-n} \bmod 23$.
 - (d) Let $h = 3$. Compute List 2: $h, h \cdot u, h \cdot u^2, \dots, h \cdot u^n \bmod 23$.
 - (e) Use the information above to finish the baby-step giant-step algorithm and find x such that $5^x \equiv 3 \bmod 23$.
13. Prove that Shank's baby-step giant-step algorithm works, i.e., show that (i) there is always a coincidence in List 1 and List 2, and (ii) that $x = i + j \cdot n$ is a number such that $g^x \equiv h \bmod p$.
14. Let $p = 17389$, let $g = 9704$, and let $h = 13896$. Use Shank's algorithm to find x such that $g^x \equiv h \bmod p$.

15. Let $p = 73$, let $E : y^2 = x^3 + 8x + 7$ over $\mathbb{F}_p$, and let $P = (32, 53)$ and $Q = (35, 47)$. Use a probabilistic collision algorithm to find n such that $n \cdot P = Q$.
16. Let $p = 73$, let $E : y^2 = x^3 + 8x + 7$ over $\mathbb{F}_p$. Determine the group structure of $E(\mathbb{F}_{73})$.
17. Alice and Bob want to communicate using a Diffie–Hellman key exchange first, using $p = 13$ and $g = 6$.
- Alice picks $a = 2$. Compute $A \equiv g^a \pmod{p}$.
 - Alice receives $B = 2$ from Bob. Compute the secret key produced by the Diffie–Hellman key exchange that Alice and Bob will share.
 - Your name is Eve and you are *very nosy*. You intercept some messages from Alice and Bob who are setting up a Diffie–Hellman key exchange. You now know that $p = 13$, $g = 6$, and intercept $A = 3$ from Alice, and $B = 5$ from Bob. Use this information to generate the secret key that Alice and Bob are planning to share. (Note: it is not the same key that they were sharing in part (b) of this problem).
18. The following is a table of some powers modulo 17:

$x \pmod{17}$	x^2	x^3	x^4	x^5	x^6	x^7	x^8	x^9	x^{10}	x^{11}	x^{12}	x^{13}	x^{14}	x^{15}	x^{16}
4	16	13	1	4	16	13	1	4	16	13	1	4	16	13	1
6	2	12	4	7	8	14	16	11	15	5	13	10	9	3	1
9	13	15	16	8	4	2	1	9	13	15	16	8	4	2	1
10	15	14	4	6	9	5	16	7	2	3	13	11	8	12	1
13	16	4	1	13	16	4	1	13	16	4	1	13	16	4	1
15	4	9	16	2	13	8	1	15	4	9	16	2	13	8	1

- Princess Leia is expecting communications from the rebel alliance, and sets up an Elgamal public cryptography system, with $p = 17$ and $g = 10$. She chooses $a = 5$ and publishes p , g , and $A \equiv g^a \pmod{p}$.
 - Compute A .
 - Luke wants to send the message $m = 3$ to Leia. Explain what steps Luke has to do to send the message, and produce an encrypted version (c_1, c_2) of the message.
 - Leia receives an encrypted message $(15, 9)$ from Chewbacca. Use the Elgamal method to decipher Chewbacca's message.
 - Governor Tarkin intercepts a rebel transmission $(13, 2)$, which has been encoded with an Elgamal cryptosystem (different from the one above). He knows that $p = 17$, $g = 6$, and $A = 9$. Decipher the transmission.
19. The number $p = 179424673$ is prime and 15 is a primitive root mod p . Find x such that $15^x \equiv 2 \pmod{p}$.